

CONDITIONALS AND LOOPS

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Date ___/___/___

Conditions:

example.

i/p $\rightarrow$ a, b $a > b \rightarrow$ "Answer is A" $a < b \rightarrow$ "Answer is B"

Pseudo code

```

if (a > b) {
    cout << "Answer is A"
}
else {
    print("Answer is B")
}

```

code:

```

#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    if (n > 0) {
        cout << "n is +ve" << endl;
    }
    else {
        cout << "n is -ve" << endl;
    }
}

```

cin can't read
 " " space
 "\t" tab
 "\n" enter

Code:

```

#include <iostream>
using namespace std;
int main() {
    int a, b;
    cin >> a >> b;
    if (a > b) {
        cout << "a is greater" << endl;
    }
    else {
        cout << "b is greater" << endl;
    }
}

```

example find input number is +ve -ve or zero.

```
#include <iostream>
using namespace std;
int main() {
    int a;
    cout << "Enter a " << endl;
    cin >> a;
    if (a > 0) {
        cout << "a is +ve" << endl;
    }
    elseif (a < 0) {
        cout << "a is -ve" << endl;
    }
    else {
        cout << "a is zero" << endl;
    }
}
```

example input char ch.

if in between 'a' - 'z' print lower case
 else if in between 'A' - 'Z' print upper case
 else in between 0 - 9 print numeric

```
#include <iostream>
using namespace std;
int main() {
    char c;
    cout << "Enter a character: ";
    cin >> c;
```

Date ___/___/___

```
if (c >= 'a' && c <= 'z') {
    cout << "Lower case";
}
else if (c >= 'A' && c <= 'Z') {
    cout << "Upper case";
}
else {
    cout << "numeric";
}
}
```

Looping

- while
- for
- do while

while loop

```
while ( ) {
    //
}
```

जब तक यह True

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example print the value from 1 to n

```
#include <iostream>
using namespace std;
int main () {
    int a;
    cin >> a;
    int i = 1;
    while (i <= a) {
        cout << i << endl;
        i = i + 1;
    }
}
```

example - Sum of the number from 1 to n

```
#include <iostream>
using namespace std;
int main() {
    int a;
    int sum = 0;
    cin >> a;
    int i = 1;
    while (i <= a) {
        sum = sum + i;
        i = i + 1;
    }
    cout << sum;
}
```

H.W. example

find sum of all even number.

```
#include <iostream>
using namespace std;
int main() {
    int a;
    int sum = 0;
    cin >> a;
    int i = 0;
    while (i <= a) {
        sum = sum + i;
        i = i + 2;
    }
    cout << sum;
}
```

(2+4+6+8+12) To

H.V. ~~example~~ Fahrenheit to Celsius conversion

```
#include <iostream>
using namespace std;
int main(){
    int f;
    cin >> f;
    int c;
    c = (f-32)*5/9;
    cout << c;
}
```

example

check number n is prime or not.

```
#include <iostream>
using namespace std;
int main(){
    int a;
    cin >> a;
    int i=2;
    while (i<a){
        if (a%i==0){
            cout << "not prime" << endl;
        }
        i = i+1;
    }
}
```

output: 8

not prime

not prime